

Technical companion to

*Long runs and rare patterns of a random
completely multiplicative function*

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Abstract

This companion records the detailed inputs and verifications used in the article. The square-product argument is given once, in the article, in a macroscopic geometry that also contains the dyadic and bounded-ratio cases. Here we verify the growing-degree Runge calculation and the bounded-height Pell count, the prime-cutoff estimates, scalar smoothing, aggregated and labelled marked comparisons, and conditional-kernel consequences. The boundary verification uses a simple incidence graph and a global count of square divisibilities to obtain a rank surplus $28271/332640$ and the displayed relative exponent $1/12$. The signed marked-field comparison retains the sign of each run. The article's capped value-relation estimate yields joint laws for growing families of prescribed words; this companion records the marginal-ceiling calculation, local ranks and tail estimates used in those transfers. Appendix letters refer to this companion; numbered sections and unqualified numbered statements refer to the article.

Reading convention. The two documents form one proof package. Each appendix below is called from a specific step in the article. Bibliographic numbering is local; cross-document theorem numbers are generated from the two LaTeX auxiliary files. No formal certification of these texts is claimed. All logarithms without an indicated base are natural.

Artificial-intelligence assistance. The drafting of this companion and work on the associated formalization received assistance from language-model systems developed by Anthropic and OpenAI. The [declaration at the end of the article](#) identifies the models, their roles and the author's responsibility.

Contents

External inputs and internal deductions	2
How to use this companion	3
A Runge coefficients and uniform Diophantine inputs	4
B Prime cutoffs and scalar smoothing	7
C Marked fields and spatial aggregation	10
D Conditional laws and the Poisson target	16
E The microscopic boundary and relative factorization	19

External inputs and internal deductions

This register identifies the non-elementary inputs, their precise role, and the calculations proved in this package. References are local to this companion; statement links lead to the relevant document. Classical finite-dimensional linear algebra, conditioning and elementary inequalities are not separate imported hypotheses.

Input and source	Form consumed	Use and internal deduction
Prime number theorem; standard prime sums [14]	$\pi(t) \sim t/\log t$; elementary weighted consequences	Proposition B.1 proves the moving weighted sum, its uniform error and both cutoff balances.
Finite-dependency Poisson approximation [1, 2]	Scalar and process theorems; $b_3 = 0$ for an exact graph	Theorems 4.1, 5.1 , and 5.6 ; the supremum convention for variation is half the sources' norm.
Scalar Stein factors [8, (2.15)]	Zeroth and first differences, of orders $\lambda^{-1/2}$ and λ^{-1}	Lemma B.2 proves the exceptional-set decomposition; no Malliavin input is used.
Directional Poisson factor [4, p. 179, Lemmas 2–3]	Weighted quadratic Hessian bound and $ \Delta_{ij}g \leq 1$, for the same solution	Equation (C.3) and Theorem C.1; Hessian conjugation, independent Poisson filling and geometric mark summation are explicit.
Quadratic ideals and units [7]; divisors [10]	Ideal factorization, unit orbits and $\tau(n) = \exp(O(\log n/\log \log n))$	Lemma A.2 proves the uniform bounded-height Pell count. Split-product localization is internal.
Laishram–Shorey [9, eq. (14), p. 331]	$\omega(\prod_{j < B}(n+j)) \geq (2-\varepsilon)\pi(B)$, uniformly for $n > B$	Appendix E.2 records the quantifiers; the private-coordinate and medium-prime rank arguments are internal.
Balasubramanian–Shorey [3], as recorded in [13, pp. 55–56]	Square-product exclusion with $P^+(b) \leq k$ and the stated height/density conditions	Lemma E.3 gives the substitutions and contrapositive. No size bound on the smooth coefficient is added.
Berry's normal approximation bound [5]	Third-absolute-moment normal approximation	Appendix D uses the explicit Poisson specialization; the marked comparison and joint limiting independence are proved here.
Binary marker construction [6]	A classical cross-bifix-free family	Corollary 5.4 proves the particular binary construction and its size bound. This illustrative application is not an input to the arithmetic theorem.

Runge's split-product estimate, the Hamming-ball argument, the raw and capped two-window profiles, the simple-incidence rank lemma and the relative bulk normalization are proved in the article–companion pair. Evertse–Silverman is contextual only; the separate process article is not a proof dependency. The table concerns the mathematical inputs to the written arguments. In particular the directional estimate is obtained directly from Barbour's two lemmas, while the arithmetic comparison and Poisson filling are proved here.

How to use this companion

This companion and the article form a single proof package. The article gives the objects, main arguments and conclusions; the entries below identify the calculation behind each excursion and the point at which the main argument resumes. Unqualified numbers refer to the article; lettered numbers refer to this companion.

Entry from the article	Calculation here	Return to the article
Lemma 2.3	Coefficients, integrality and the growing-degree remainder in Lemma A.1.	Weighted defects, proposition 2.4 .
Lemmas 3.11 and 3.12	Uniform ideal and unit counts, then localization to two square classes in A.2–A.3.	Bounded components and the deep-core estimates.
Proposition 4.2 and Theorem 4.3	Weighted prime sums, degree bounds and scalar retention in B.1–B.3.	Scalar laws or the fields of Section 5.
Theorem 5.1	Marginal-cap substitutions and exponent bookkeeping in C.1.	Dictionary examples and consequences.
Theorem 5.8(ii)	The two Stein bounds, Poisson filling and the single tail event in C.3–C.4.	The complete staircase and resolved kernels.
Theorem 6.3	Poisson atom estimates and the resolution cost in D.1–D.2.	Conditional laws in Section 6.
Theorem 7.1	Prime incidences, quotient rank and the post-quadratic bound in E.1–E.5.	The microscopic mass and the prefix law.
Theorems 7.7 and 7.9	The normalized error table, finite-cylinder clock and affine conditioning in E.6–E.7.	The two-source comparison and its affine variant.

Support conventions. The same symbols have different roles in different families of events. For a run of length at least L , the system has L relative equations on $B = L + 1$ vertices. An absolute word of length B fixes B values. An exact run with excess e uses $L + e + 1$ relative equations on $L + e + 2$ vertices; recording its sign prescribes all these values. After truncating at $e \leq E$, the common maximal support has $Q + 1$ vertices, where $Q = L + E + 1$. In contrast, $Q_B = 2^B$ is the exponential parameter in the arithmetic relation profile.

Two different summations. For a spatially labelled field, marks are summed without weights before the pair estimate, leaving the relation profile at the base length. For an aggregated vector, the directional Stein weights are inserted first, and the maximal-length profile is required. Section C.3 keeps this distinction explicit before any asymptotic absorption.

A Runge coefficients and uniform Diophantine inputs

This appendix supplies the growing-degree Runge calculation and the bounded-height Pell argument used in the article. The two types of uniformity are different: the Runge degree is allowed to grow, whereas a component's degree and its polynomial-height exponent are fixed before the ambient scale. No height-free integral-point estimate is needed.

A.1 The split-product calculation

Lemma A.1 (Coefficient-level Runge estimate). *Under the hypotheses of Lemma 2.3, including $U \in \mathbb{Z}$, the bound $U \leq (C_0 d R)^{2d}$ holds with an absolute constant C_0 .*

Proof. The γ_ν being distinct and $d \geq 2$, we have $R \geq 1$ and G is not the square of a polynomial. Set

$$F(z) = \prod_{\nu=1}^d (1 + \gamma_\nu z)^{1/2} = \sum_{m \geq 0} c_m z^m,$$

where each square root is the analytic branch equal to 1 at 0, and

$$P(T) = \sum_{m=0}^k c_m T^{k-m}.$$

For $m \leq k$, the identity

$$\binom{1/2}{j} = (-1)^{j-1} \frac{\text{Cat}_{j-1}}{2^{2j-1}} \quad (j \geq 1)$$

implies $2^{2j} \binom{1/2}{j} \in \mathbb{Z}$ and $|\binom{1/2}{j}| \leq 1$. More explicitly,

$$c_m = \sum_{j_1 + \dots + j_d = m} \prod_{\nu=1}^d \binom{1/2}{j_\nu} \gamma_\nu^{j_\nu}.$$

The number of weak compositions of m into d parts is $\binom{m+d-1}{d-1} \leq 2^{m+d}$. Since $m \leq k = d/2$ and $|\gamma_\nu| \leq R$, we obtain simultaneously

$$2^d c_m \in \mathbb{Z}, \quad |c_m| \leq 2^{m+d} R^m \leq (8R)^d, \quad 0 \leq m \leq k. \quad (\text{A.1})$$

In particular $2^d P \in \mathbb{Z}[T]$, uniformly as d grows with the ambient scale.

If $2R \leq U \leq 4R$, the conclusion is immediate after enlarging C_0 . Assume therefore $U > 4R$. On the circle $|z| = (2R)^{-1}$,

$$|F(z)| \leq (3/2)^{d/2} =: M_d.$$

Cauchy's formula gives $|c_m| \leq M_d (2R)^m$. Since $2R/U < 1/2$,

$$\begin{aligned} |\sqrt{G(U)} - P(U)| &= U^k \left| \sum_{m \geq k+1} c_m U^{-m} \right| \\ &\leq U^k M_d \sum_{m \geq k+1} (2R/U)^m \\ &\leq \frac{2M_d (2R)^{k+1}}{U} \leq \frac{(C_1 d R)^d}{U}. \end{aligned} \quad (\text{A.2})$$

If the difference is nonzero, $\sqrt{G(U)}$ is an integer and, by (A.1), $P(U) \in 2^{-d}\mathbb{Z}$. It is therefore at least 2^{-d} in absolute value. Inequality (A.2) then implies

$$U \leq 2^d (C_1 d R)^d \leq (C_0 d R)^{2d}.$$

Assume finally $\sqrt{G(U)} = P(U)$. The polynomial

$$Q(T) = 2^{2d}G(T) - (2^d P(T))^2$$

has integer coefficients. It is nonzero, since G is not a square. The expansion at infinity gives $\sqrt{G(T)} - P(T) = O(T^{-1})$, whence

$$\deg Q \leq k - 1.$$

The coefficient of T^r in G is a symmetric polynomial containing at most $\binom{d}{r} \leq 2^d$ monomials, each of modulus at most R^d after the uniform upper bound $R \geq 1$. By (A.1), every coefficient of $2^d P$ is at most $(16R)^d$ in modulus. A coefficient of $(2^d P)^2$ is the sum of at most $d + 1$ products of two such coefficients. Consequently

$$H(2^{2d}G) \leq (8R)^{2d}, \quad H((2^d P)^2) \leq (d + 1)(16R)^{2d},$$

and therefore, for an absolute constant C_2 ,

$$H(Q) \leq (C_2 d R)^{2d}.$$

This estimate is uniform in d ; it hides no step where d would be fixed. An integer zero of a nonzero integer polynomial has modulus at most $1 + H(Q)$, by the elementary Cauchy bound. After enlarging C_0 , we again obtain (2.2). $\square$

The integrality of U is used precisely in the separation by 2^{-d} . Without it the statement is false: for $G(T) = T(T + 1)$ the real numbers $U_m = (\sqrt{1 + 4m^2} - 1)/2$ tend to infinity and satisfy $G(U_m) = m^2$.

A.2 Pell equations with varying coefficients

Lemma A.2 (Uniform Pell count in a polynomial range). *Fix $K_0 > 0$. Let A, C be positive squarefree integers such that A/C is not a rational square, and let $e \neq 0$. Assume*

$$A, C, |e| \leq M^{K_0}.$$

The number of integer solutions (z, w) of

$$Az^2 - Cw^2 = e, \quad |z|, |w| \leq M^{K_0},$$

is

$$\exp\left(O_{K_0}\left(\frac{\log M}{\log \log M}\right)\right). \quad (\text{A.3})$$

Proof. Set $g = (A, C)$, $A = gA_0$, $C = gC_0$. Since A, C are squarefree, A_0, C_0 are squarefree, coprime, and their prime supports are disjoint. If a solution exists, $g \mid e$; write $e = ge_0$. After multiplying by A_0 and substituting $X = A_0 z$, the equation becomes

$$X^2 - Dw^2 = m_0, \quad D = A_0 C_0, \quad m_0 = A_0 e_0. \quad (\text{A.4})$$

The product D is squarefree. The hypothesis on A/C gives $D > 1$.

In the real quadratic field $K_D = \mathbb{Q}(\sqrt{D})$, using the standard ideal-and-unit description of quadratic orders [7], the element

$$\alpha = X + w\sqrt{D}$$

belongs to the ring of integers $\mathcal{O}_{K_D}$, since $\sqrt{D}$ is an algebraic integer. Its norm is m_0 . Consequently

$$(\alpha)(\bar{\alpha}) = (m_0),$$

and the integral ideal (α) divides the ideal (m_0) . For each rational prime $p^a \parallel m_0$, the number of choices for the part above p is at most $(a+1)^2$: it equals $(a+1)^2$ if p splits, $a+1$ if it is inert, and $2a+1 \leq (a+1)^2$ if it ramifies, since then $(p) = \mathfrak{p}^2$. The number of ideal divisors of (m_0) is therefore at most

$$\prod_{p^a \parallel m_0} (a+1)^2 = \tau(|m_0|)^2.$$

Only the principal ideals $\mathfrak{a} = (\alpha)$ of absolute norm $|m_0|$ that admit a generator of norm exactly m_0 can come from a solution of (A.4). Fix such an ideal and a generator α_0 of norm m_0 . Then

$$|\sigma_1(\alpha_0)\sigma_2(\alpha_0)| = |m_0|.$$

All relevant generators are among

$$\pm \alpha_0 \varepsilon_D^n, \quad n \in \mathbb{Z},$$

where $\varepsilon_D > 1$ generates the free part of the unit group. If a unit of norm -1 exists, the norm- m_0 condition selects only one parity class of n , which cannot increase the number of solutions.

The bounds on A, C, z, w give, for the two real embeddings σ_1, σ_2 ,

$$|\sigma_j(\alpha)| \leq H := M^{O_{K_0}(1)}.$$

The inequalities

$$|\sigma_1(\alpha_0)|\varepsilon_D^n \leq H, \quad |\sigma_2(\alpha_0)|\varepsilon_D^{-n} \leq H$$

define an interval of integers n of length at most

$$\frac{2 \log H - \log |m_0|}{\log \varepsilon_D} + O(1) = O_{K_0}(\log M),$$

since every real quadratic unit $\varepsilon > 1$ satisfies $\varepsilon \geq (1 + \sqrt{5})/2$. Indeed its conjugate is $\pm \varepsilon^{-1}$ and its trace is an integer. For norm -1 , this gives $\varepsilon - \varepsilon^{-1} \geq 1$; for norm $+1$, $\varepsilon + \varepsilon^{-1} \geq 3$. Both give the asserted lower bound. Thus the number of solutions is at most

$$O_{K_0}(\tau(|m_0|)^2 \log M).$$

Since $|m_0| \leq M^{O_{K_0}(1)}$, the standard explicit divisor bound [10] gives

$$\tau(|m_0|)^2 \log M = \exp\left(O_{K_0}\left(\frac{\log M}{\log \log M}\right)\right),$$

which proves (A.3); the sign of m_0 does not affect the argument. $\square$

The exponent K_0 is fixed in every use. There is no appeal to a uniform lower bound for a regulator that deteriorates with the discriminant: the elementary absolute lower bound on a real quadratic fundamental unit already controls the number of generators in each ideal class.

A.3 Square-class localization in bounded height

In Lemma 3.12, the decisive set is $S = \{p : p \mid e \prod_{i < j} (h_i - h_j)\}$. For a solution, a prime outside S can divide only one shifted factor; its valuation there is therefore even. We enumerate the squarefree parts of only the first two factors, not of all d factors. There are at most $4^{|S|}$ pairs, and the two-factor equation has a nonzero right-hand side because $h_1 \neq h_2$.

Here is an explicit check on the order of parameters. Fix d, K first. For $M \geq 2$,

$$\log |e\Delta| \leq K \log M + \binom{d}{2} (K \log M + \log 2) = O_{d,K}(\log M).$$

Thus $4^{|S|} = \exp(O_{d,K}(\log M / \log \log M))$. Independently, every retained squarefree coefficient is at most $2M^K$, not an unrestricted product of primes in S . The coefficients, right-hand side and square variables of its Pell equation are therefore all covered by the fixed exponent $K + 1$ (increasing the ambient threshold if needed). The Pell constant does not depend on S , the offsets or the chosen pair. When the squarefree coefficients agree, divisor factorization treats the case that the nonsquare Pell lemma deliberately excludes.

For a component, $d = |J| \leq K_0$ and the normalizing coefficient is the squarefree part of $d_C \prod_{i \in I} (x + i)$. The smooth coefficient $d_C \leq \exp(O(B))$ and $|I| \leq K_0$ put this integer in one fixed polynomial range. This supplies exactly the uniformity consumed by Lemma 3.14, even when the two starts have different scales.

A.4 Why a single rational unit is not contracted

Two units force the reduced channel height to be at most L . One unit does not. For example, take $N = 2^{100}$, $L = 100$, $a = 103$, $b = 102$ and $t = 12427947061061072563693168687$, with $x = bt$, $y = at$. Both starts lie in $[N, 2N)$, and $103i = 102j$ has only $(i, j) = (0, 0)$ in $\{-1, \dots, 99\}^2$. Yet $103 > B = 101$, and $t \equiv 100 \pmod{103}$ gives $v_{103}(x) = 0$, $v_{103}(y) = 1$. No vertex of the first window is divisible by 103, so the second occurrence is pinned rather than isolated with the first. This is why the article defines $\sigma = 0$, $D^\# = D$ and $c^\# = c$ when $m \leq 1$. The aligned deep-core proof removes the at most two components meeting a single unit directly; it never assigns that unit a false private-kernel identity.

A.5 Uniformity across the master interval

All vertices are $O(M)$ and at least $M^\delta - 1$. Squarefree B -smooth coefficients have size $M^{O(1)}$, while a fixed number of offsets and smooth choices contributes $M^{o(1)}$. On a larger-start terminal slice $X \leq \max(x, y) < 2X$, $B/\log X$ remains in a fixed compact interval depending only on $(\delta, \beta_-, \beta_+)$. Thus the Pell estimates can be applied at scale X there, even when the other start is smaller. The proof never identifies a macroscopic cross-block pair with an intra-block pair, and never applies a fixed-ratio theorem with a growing ratio.

B Prime cutoffs and scalar smoothing

This appendix verifies the free cutoff before specializing it to either saddle. The distinction matters: the hard route deletes all exceptional starts, whereas the soft scalar route leaves them in the Stein equation.

B.1 A uniform weighted prime sum

Proposition B.1 (Free Rankin calculation and both saddles). *The bounds and expansions in [proposition 4.2](#) hold uniformly for w in a fixed multiplicative neighbourhood of $\sqrt{H \log H}$, where $H = \log N$.*

Proof. Every Y -defective integer has the unique form $n = a^2 s$ with s squarefree and Y -smooth. Therefore, for $0 < \zeta < 1/2$,

$$\frac{A(X, Y)}{X} \leq X^{-\zeta} \prod_{p \leq Y} (1 + p^{-1+\zeta}). \quad (\text{B.1})$$

Indeed sum $\sqrt{X/s}$ over $s \leq X$ and use $s^{-1/2} \leq X^{1/2-\zeta} s^{-1+\zeta}$. Put $\nu = H/w$, choose the larger solution $u \geq 1$ of $e^u/u = \nu$, and set $\zeta = u/w$. Uniformly in the stated range,

$$u \sim \frac{1}{2} \log H, \quad \nu \asymp \sqrt{H/\log H}, \quad \zeta \asymp \sqrt{\log H/H}, \quad \log H = o(\nu).$$

For X/N in a fixed compact subset of $(0, \infty)$, $\zeta \log X = u\nu + o(\nu)$.

We require the prime number theorem inside the weighted Euler product, not merely an upper bound with an unspecified constant. Let $R(t) = \pi(t) - \text{Li}(t)$. Given $\epsilon > 0$, choose a fixed lower split point beyond which $|R(t)| \leq \epsilon t / \log t$. Stieltjes integration by parts gives, uniformly in w ,

$$\begin{aligned} \sum_{p \leq e^w} p^{-1+\zeta} &= \int_2^{e^w} \frac{t^{-1+\zeta}}{\log t} dt + O_\epsilon(1) + O\left(\epsilon \frac{e^u}{w} + \epsilon \int_2^{e^w} \frac{t^{-1+\zeta}}{\log t} dt\right) \\ &= \text{Ei}(u) + o(\nu). \end{aligned}$$

For the second equality first let $N \rightarrow \infty$ and then $\epsilon \downarrow 0$. The substitution $z = \zeta \log t$ costs only $O(\log(1/\zeta)) = o(\nu)$ at the lower endpoint, and $\text{Ei}(u) \asymp e^u/u = \nu$. Also $\sum_p p^{-2+2\zeta} = O(1)$, so logarithms of the Euler factors give

$$\sum_{p \leq e^w} \log(1 + p^{-1+\zeta}) = \text{Ei}(u) + o(\nu). \quad (\text{B.2})$$

Floors in $Y = \lfloor e^w \rfloor$ alter these estimates by a negligible quantity. Equations (B.1)–(B.2) yield

$$\log(A(X, Y)/X) \leq -\nu u + \text{Ei}(u) + o(\nu) = -D(\nu) + o(\nu).$$

A bad integer contaminates at most B windows; $\log B = o(\nu)$ absorbs this factor. The graph-edge estimate in the article gives $E_Y/N^2 \leq e^{-w+o(\nu)} + N^{-1+o(1)}$. A window uses at most CBH/w large primes, and each prime meets at most $CB(N/Y + 1)$ starts. Hence

$$\Delta_Y \ll B^2(H/w)(N/e^w + 1), \quad \Delta_Y/N \leq e^{-w+o(\nu)} + N^{-1+o(1)}. \quad (\text{B.3})$$

This maximum-degree bound applies to the graph on all starts, including exceptional ones.

For a unified expansion write $V_a = aD(H/V_a)$ for $a = 1$ or 2 , $\nu_a = H/V_a$, and $u_a = u(\nu_a)$. Since $D' = u > 0$, the defining function decreases strictly in V_a . At zero its sign is positive; at H/e its sign is negative for sufficiently large H . Thus the root is unique. The parametric identities are

$$\nu_a = e^{u_a}/u_a, \quad V_a = a(e^{u_a} - \text{Ei}(u_a)).$$

Using $\text{Ei}(u) = e^u u^{-1}(1 + u^{-1} + O(u^{-2}))$ gives

$$\log H = 2u_a - \log u_a + \log a + O(u_a^{-1}), \quad u_a = \frac{1}{2}(\log H + \log \log H - \log(2a)) + o(1),$$

and $V_a^2/H = a(u_a - 1 + O(u_a^{-1}))$. Substitution gives (4.6) and (4.7). $\square$

The second-order constants belong to this PNT main-term optimization. They are not matching lower bounds for the probability distance.

B.2 A finite soft-exception Stein lemma

Lemma B.2 (Retention of exceptional indicators). *Let a finite indicator family $(I_\alpha)_{\alpha \in \mathcal{I}}$ have an exact dependency graph conditionally on $\mathcal{F}$. Split $\mathcal{I} = \mathcal{G} \sqcup \mathcal{X}$ deterministically, and assume $\mathbb{E}(I_\alpha | \mathcal{F}) = p$ for every $\alpha \in \mathcal{G}$. Put $W = \sum I_\alpha$, $\lambda = |\mathcal{I}|p$, $\lambda_{\mathcal{X}} = |\mathcal{X}|p$ and $W_{\mathcal{X}} = \sum_{\alpha \in \mathcal{X}} I_\alpha$. Let $p_\beta(\omega) = \mathbb{E}(I_\beta | \mathcal{F})(\omega)$. The mean conditional Poisson distance is at most a constant times*

$$(1 \wedge \lambda^{-1}) \left[|\mathcal{G}|p^2 + p \sum_{\alpha \in \mathcal{G}} \sum_{\beta \in B_\alpha \setminus \{\alpha\}} \mathbb{E}I_\beta + \sum_{\alpha \in \mathcal{G}} \sum_{\beta \in B_\alpha \setminus \{\alpha\}} \mathbb{E}(I_\alpha I_\beta) \right] + (1 \wedge \lambda^{-1/2})(\lambda_{\mathcal{X}} + \mathbb{E}W_{\mathcal{X}}). \quad (\text{B.4})$$

The two minimum factors are interpreted as one if $\lambda = 0$. No marginal assumption is made for exceptional neighbours.

Proof. If $\lambda = 0$, all good indicators vanish almost surely and $d_{\text{TV}}(\mathcal{L}(W | \mathcal{F}), \delta_0) \leq \mathbb{E}(W_{\mathcal{X}} | \mathcal{F})$; averaging proves the claim. For $\lambda > 0$ and a target test set $A \subset \mathbb{N}_0$, let $f = f_{\lambda, A}$ solve $\lambda f(k+1) - kf(k) = \mathbf{1}_A(k) - \text{Pois}(\lambda)(A)$. The standard factors are

$$\|f\|_\infty \leq 1 \wedge \sqrt{2/(e\lambda)}, \quad \|\Delta f\|_\infty \leq (1 - e^{-\lambda})/\lambda \leq 1 \wedge \lambda^{-1}$$

[8, (2.15)]; see also [12, (2.4)]. For $\alpha \in \mathcal{G}$ set $U_\alpha = \sum_{\beta \in B_\alpha \setminus \{\alpha\}} I_\beta$ and $V_\alpha = W - I_\alpha - U_\alpha$. On each fibre I_α is independent of V_α . Add and subtract $p\mathbb{E}[f(V_\alpha + 1) | \mathcal{F}]$. Telescoping gives

$$\begin{aligned} & |p\mathbb{E}[f(W+1) | \mathcal{F}] - \mathbb{E}[I_\alpha f(W) | \mathcal{F}]| \\ & \leq \|\Delta f\|_\infty \left[p^2 + p \sum_{\beta \in B_\alpha \setminus \{\alpha\}} p_\beta + \sum_{\beta \in B_\alpha \setminus \{\alpha\}} \mathbb{E}(I_\alpha I_\beta | \mathcal{F}) \right]. \end{aligned}$$

The remaining term is bounded directly by

$$|\mathbb{E}[\lambda_{\mathcal{X}} f(W+1) - W_{\mathcal{X}} f(W) | \mathcal{F}]| \leq \|f\|_\infty (\lambda_{\mathcal{X}} + \mathbb{E}[W_{\mathcal{X}} | \mathcal{F}]).$$

The bound is uniform in the test set, so taking the supremum on each fibre and then averaging proves (B.4). $\square$

B.3 Applying the lemma at a movable cutoff

Take $\mathcal{X} = D_Y$, $p = 2^{-L}$ and $\lambda = Np > 0$. Set $\ell = \log^+ \lambda$ and $K = \max(1, \lambda)$. The elementary inequalities

$$(1 \wedge \lambda^{-1})\lambda^2 \leq K, \quad (1 \wedge \lambda^{-1})(\lambda^2 + \lambda) \leq 2K, \quad (1 \wedge \lambda^{-1/2})\lambda \leq \sqrt{K}$$

hold on both sides of intensity one. For neighbour marginals use the maximum degree of the graph on all sites:

$$p \sum_{x \in \mathcal{G}} \sum_{y \in B_x \setminus \{x\}} \mathbb{E} J_{y,L} \leq p \Delta_Y \mathbb{E} Z_{N,L} \leq p \Delta_Y (\lambda + p \mathcal{M}_B).$$

The first-difference factor makes this at most $K(\Delta_Y/N)(1 + N^{-1/2+o(1)})$. The true marginals of exceptional neighbours have not been replaced by p . For the joint sum, overlapping starts contribute zero and the local bound treats touching pairs; the separated pairs give

$$\sum_{x \in \mathcal{G}} \sum_{y \in B_x \setminus \{x\}} \mathbb{P}(J_{x,L} J_{y,L} = 1) \ll p^2 (N^{1+o(1)} + N \Delta_Y + R_2(N, L)).$$

Since $R_2/N^2 \ll N^{-1/3+\varepsilon/4}(1 + \lambda^{-1})$, its contribution, after the Stein factor, is at most $KN^{-1/3+\varepsilon/2}$ after enlarging the threshold. The diagonal and local polynomial costs are smaller. Finally the exceptional zeroth-order term is at most

$$C\sqrt{K} \left(\frac{\mathcal{M}_B}{N} + \frac{|D_Y|}{N} \right) \leq C e^{\ell/2 - D(H/w) + o(H/w)} + KN^{-1/3+\varepsilon/2}.$$

Together with (B.3), these estimates give (4.10) for all intensities.

At $w = \widehat{V}_N$, let $\nu = H/w$ and choose a common $\eta_N \geq 0$, $\eta_N \rightarrow 0$, dominating the displayed remainder terms. Put

$$q = e^{-(w-\ell)/2 + \eta_N \nu}, \quad r = e^{\ell - w + \eta_N \nu}, \quad P = N^{-1/3+\varepsilon/2}.$$

Up to an absolute constant, the error is at most $q + r + KP$. If $q + N^{-1/3+\varepsilon} \geq 1$, the trivial total-variation bound suffices. Otherwise $q < 1$ implies $\ell \leq w$ and $r \leq q$. Since $w = o(\log N)$, eventually $K \leq e^w \leq N^{\varepsilon/2}$, so $KP \leq N^{-1/3+\varepsilon}$. This proves the soft minimum bound, including $\lambda < 1$, at the same cutoff and on the full $\mathcal{F}_Y$. All thresholds depend only on the fixed band and the error margin, not on its length parameter. The numerical constants in a coarser ledger do not affect the claimed asymptotic bound.

All terms admit their conditional versions: integrate the fibrewise bound over $C \in \mathcal{F}_Y$ and divide by $\mathbb{P}(C)$. A single factor e^I multiplies the ledger, where $I = -\log \mathbb{P}(C)$; the leading soft exponent consequently has sufficient budget $2I + \log^+ \lambda < \widehat{V}$. This argument applies only to events measurable in the chosen prime field. It does not turn arbitrary post-selection on a run statistic into an arithmetic conditioning event.

C Marked fields and spatial aggregation

The distinction between a labelled field and its aggregate is already visible in a finite family. This appendix gives the finite inequalities first; all passages to countably many marks are then controlled by a single tail event.

C.1 The marginal ceiling for dictionaries

The marginal-ceiling calculation in the proof of Theorem 5.1 gives the finite inequality

$$\mathbb{E} \Delta_{\mathcal{W}} \ll a(\mathcal{M}_B^{\text{val}} + |D_Y^{\text{val}}|) + a^2(NB + E_Y + R_2^{\text{val}, [1/a]}(N, B)) + \Lambda \Omega(\mathcal{W}),$$

where $a = m2^{-B}$ and $\Lambda = Na$. The comparison retains the full site–word index set; only the error terms have been summed over words. We verify here the substitutions and the uniform

absorption of secondary terms. With $T = N/\Lambda$ and $Q_B = Nm/\Lambda \geq T$, the three capped-profile terms become, respectively,

$$N^{-1/3+\varepsilon} \Lambda^{11/6} m^{1/6}, \quad N^{-1/3+\varepsilon} \Lambda^{4/3} m^{2/3}, \quad N^{-1/3+\varepsilon} \Lambda.$$

These substitutions are exact because $a = \Lambda/N$ and $Q_B = Nm/\Lambda$. The restrictions $1 \leq m \leq 2^B$ also give $0 < a \leq 1$, so the arithmetic cap $T = 1/a$ is always at least one. The secondary edge contribution is absorbed in every admissible intensity regime, not only at criticality: $m \geq \Lambda/N$ implies

$$\frac{\Lambda^2}{N} \leq N^{-1/3} \Lambda^{4/3} m^{2/3}.$$

Factors $N^{o(1)}$ in these smaller terms are absorbed by fixing a smaller arithmetic error exponent first. At bounded positive Λ and $m \leq N^{1/2-\eta}$, the powers of N in the three terms, with $\varepsilon = \eta/6$, are at most

$$-\frac{1}{4}, \quad -\frac{\eta}{2}, \quad -\frac{1}{3} + \frac{\eta}{6} \leq -\frac{\eta}{2} \quad (0 < \eta < 1/2).$$

Thus all three are $O(N^{-\eta/2})$, uniformly in the dictionary. No independent random marks are added to the multiplicative sequence, and no word identity is lost when the error terms are summed.

C.2 Local pairs and the role of the maximal support

From here on, $p = 2^{-L}$, $q_e = 2^{-e-1}$, $p_e = pq_e$, $\lambda = Np$ and $0 \leq e \leq E$. The largest row length is $Q = L + E + 1$; its support has $Q + 1$ vertices. An unsigned mark of excess e imposes $L + e + 1$ relative equations; prescribing its sign fixes all $L + e + 2$ values on the same support.

A good maximal support contains no Y -defective vertex and has diameter $Q < Y$. Every exact-mark system on that support consequently has full rank above Y , with conditional marginal p_e . If two actual supports intersect, their union has diameter at most $2Q < Y$; every vertex of the union then has a prime coordinate private to that union. For two compatible exact runs the only possible intersecting cases are a common terminal/border vertex or a common terminal edge. Equivalently, for $x < y$, they satisfy $y - x = L + e + 1$ or $y - x = L + e$. The union of the two constraint trees has cycle rank at most one, so its probability is at most $2p_e p_f$.

The actual supports can also be disjoint while the *maximal* supports intersect. In that case all their private coordinates are still distinct, and the probability is $p_e p_f$. Thus the bound

$$\mathbb{P}(K_{x,e} K_{y,f} = 1 \mid \mathcal{F}_Y) \leq 2p_e p_f \quad (0 < |x - y| \leq Q) \quad (\text{C.1})$$

holds for every good pair; it does not assert that only two separations are possible among pairs of intersecting maximal supports. At the same site, different exact marks are mutually exclusive.

For $|x - y| > Q$, zero-extension of the coefficient vectors embeds the mixed relation space into the two-window space at row length Q . Therefore

$$\mathbb{P}(K_{x,e} K_{y,f} = 1) \leq p_e p_f 2^{\rho_Q(x,y)}. \quad (\text{C.2})$$

This is an averaged, not necessarily fibrewise, estimate: the small-prime phase in the affine formula can vary with the environment.

For signed marks, put $p_{e,s} = p_e/2$ and let $a = L + e$, $b = L + f$. On a good local union the

raw value vector is uniform. For $x < y$, $d = y - x$, the exact joint probabilities are

$$\mathbb{P}(K_{x,e}^s K_{y,f}^t = 1 \mid \mathcal{F}_Y) = \begin{cases} 0, & d < a, \\ 4p_{e,s}p_{f,t}\mathbf{1}_{\{t=-s\}}, & d = a, \\ 2p_{e,s}p_{f,t}\mathbf{1}_{\{t=s\}}, & d = a + 1, \\ p_{e,s}p_{f,t}, & a + 1 < d \leq Q. \end{cases}$$

The final case concerns disjoint actual supports inside intersecting maximal supports. The factor four, rather than two, is needed when signs are prescribed. After summing signs one recovers (C.1). For separated signed systems, zero-extension instead gives the bound $p_{e,s}p_{f,t}2^{\rho_{Q+1}^{\text{val}}(x,y)}$. [Proposition 3.24](#) controls its weighted sum.

C.3 The directional multivariate Stein factor

Let $\pi = \bigotimes_{i=1}^d \text{Pois}(\lambda_i)$, with $d \geq 2$, $\lambda_i > 0$, $\lambda' = \sum_i \lambda_i$ and $\mu_i = \lambda_i/\lambda'$. For a test set $A \subseteq \mathbb{N}_0^d$, use Barbour's immigration-death solution, with its sign chosen so that

$$\mathcal{A}g(z) = \mathbf{1}_A(z) - \pi(A), \quad \mathcal{A}g(z) = \sum_i \{\lambda_i \Delta_i g(z) + z_i [g(z - \mathbf{e}_i) - g(z)]\}.$$

At $z_i = 0$ the corresponding death term is zero. Write $H(z) = (\Delta_{ij}g(z))_{i,j}$ and $c(\lambda') = (1 + 2\log^+(2\lambda'))/2$. Barbour's Lemmas 2 and 3 [4, p. 179] give, for this same solution,

$$|\Delta_{ij}g(z)| \leq 1, \quad |\alpha^T H(z) \alpha| \leq c(\lambda') \sum_i \frac{\alpha_i^2}{\lambda_i}.$$

The weighted bound, stated there for integer directions, extends to rational directions by homogeneity and to real directions by continuity. The literature input consists of the solution and these analytic bounds, not an arithmetic or aggregated total-variation estimate.

For clarity, the constant entrywise bound also follows directly from the solution's semigroup representation. Couple the processes started at z , $z + \mathbf{e}_i$, $z + \mathbf{e}_j$ and $z + \mathbf{e}_i + \mathbf{e}_j$ by adding two independent particles of rate-one exponential lifetimes. The second difference vanishes once either added particle dies; before that time the second difference of an indicator is at most two in absolute value. Its time integral is thus bounded by $\int_0^\infty 2e^{-2t} dt = 1$. The argument includes $i = j$.

To obtain the weighted entrywise bound, conjugate the symmetric matrix $H(z)$ by $D = \text{diag}(\sqrt{\lambda_i})$. The quadratic inequality gives $\|DH(z)D\|_{2 \rightarrow 2} \leq c(\lambda')$. Bounding each entry by this norm, and using the constant bound above, yields

$$|\Delta_{ij}g(z)| \leq \min \left\{ 1, \frac{1 + 2\log^+(2\lambda')}{2\lambda' \sqrt{\mu_i \mu_j}} \right\} \leq \min \left\{ 1, \frac{C(1 + \log^+(2\lambda'))}{\lambda' \sqrt{\mu_i \mu_j}} \right\}. \quad (\text{C.3})$$

The logarithmic term in multivariate Stein factors is discussed in [12]; the two bounds used above are taken directly from Barbour. For the truncated geometric marks, $S_E = \sum_{e=0}^E q_e \in [1/2, 1]$, $\lambda' = \lambda S_E$ and $\mu_e = q_e/S_E$. The second expression is thus bounded by $C(1 + \log^+(2\lambda))/(\lambda \sqrt{q_e q_f})$. The summability that removes dependence on the number of marks is

$$\sum_{e \geq 0} \sqrt{q_e} = 1 + \sqrt{2}. \quad (\text{C.4})$$

Theorem C.1 (Finite aggregated comparison). *Let $X_e = \sum_{x \in I_N} K_{x,e}$, and let $D_Y(Q)$ be the bad maximal supports. Put*

$$s(\lambda) = 1 \wedge \frac{C(1 + \log^+(2\lambda))}{\lambda}.$$

Uniformly while $L + 1$ and $Q + 1$ remain in a fixed positive logarithmic band and $2Q < Y$, one has

$$\begin{aligned} \mathbb{E} d_{\text{TV}} \left(\mathcal{L}((X_e)_{0 \leq e \leq E} \mid \mathcal{F}_Y), \bigotimes_{e=0}^E \text{Pois}(\lambda q_e) \right) \\ \ll p\mathcal{M}_B + p|D_Y(Q)| + s(\lambda)p^2(NQ + E_Y(Q) + R_2(N, Q)). \end{aligned} \quad (\text{C.5})$$

Here $\mathcal{M}_B$ is the one-window defect mass at the base length, and $E_Y(Q)$ is the ordered large-prime edge count for maximal supports.

Proof. Write $D = D_Y(Q)$ and retain the good-site counts $G_e = \sum_{x \notin D} K_{x,e}$. Independently of the arithmetic variables, introduce independent $Z_e \sim \text{Pois}(\zeta_e)$ with $\zeta_e = |D|pq_e$, and put $W = G + Z$. Removing the actual bad-site marks and adding this Poisson filling changes the law by at most

$$\sum_{x \in D} \mathbb{P}(J_{x,L} = 1) + \sum_{e=0}^E \zeta_e \ll p\mathcal{M}_B + p|D|.$$

This is an averaged conditional coupling bound. It has no factor E , since the actual marks are disjoint refinements of $J_{x,L}$ and $\sum_{e \leq E} q_e \leq 1$.

Use the Stein solution for the *full* target means λq_e . For the coordinate vector $\mathbf{e}_e$ and $\Delta_e g(w) = g(w + \mathbf{e}_e) - g(w)$, Poisson summation gives, conditionally on the good indicators and $\mathcal{F}_Y$,

$$\mathbb{E}[Z_e \Delta_e g(W - \mathbf{e}_e) \mid G, \mathcal{F}_Y] = \zeta_e \mathbb{E}[\Delta_e g(W) \mid G, \mathcal{F}_Y].$$

Indeed $z\mathbb{P}(Z_e = z) = \zeta_e \mathbb{P}(Z_e = z - 1)$ for $z \geq 1$; the term with $Z_e = 0$ is zero. Thus the filling terms cancel the ζ_e part of the birth rates in

$$\sum_{e=0}^E \{\lambda q_e \Delta_e g(W) - W_e \Delta_e g(W - \mathbf{e}_e)\}.$$

Only the good indicators remain to be telescoped. The target parameters are still $\lambda' = \lambda S_E$ and $\mu_e = q_e/S_E$, so $\lambda' \sqrt{\mu_e \mu_f} = \lambda \sqrt{q_e q_f}$ exactly. No comparison of the good-site intensity with λ is needed, even if every site was removed. The independent vector Z stays in the outside count during each telescoping step, so the good-site dependency graph is unchanged. For good sites, apply the multivariate dependency-graph Stein argument [4, 12] to the *counts by mark*, not to counts by site and mark. On the first-difference telescoping terms each pair of types contributes its joint or product mass multiplied by $|\Delta_{ef}g|$. The product masses are $p_e p_f$. For local pairs use (C.1); for separated pairs use (C.2) after averaging. Decompose $2^{\rho_Q} = 1 + (2^{\rho_Q} - 1)$ and sum the second part by $R_2(N, Q)$. The diagonal and all local pairs are bounded by $O(NQp^2)$.

Using the first bound in (C.3) sums the mark weights to at most one. Using the second sums them to $\sum_{e,f} \sqrt{q_e q_f} \leq (1 + \sqrt{2})^2$. Take the better of these two estimates. This proves (C.5) when $E \geq 1$. For $E = 0$, retain both signs during the same comparison, so there are two positive target coordinates. At fixed excesses the factor does not depend on the chosen signs; summing their probabilities before the relative-pair bound again gives $R_2(N, Q)$. The signed local factor is at most four, an absolute constant. Thus the result also covers $E = 0$ without extending the literature statement to dimension one. Notice that the arithmetic argument is

used at Q , rather than merely at L : after insertion of the directional weights, the unweighted folding identity for marks is not the inequality being summed. $\square$

At $Y = Y_N^\sharp$, the first two arithmetic error sources in (C.5) are bounded by

$$\lambda e^{-V_N + o(\nu_N)} + N^{-1/2 + o(1)} \lambda, \quad \lambda(1 + \log^+(2\lambda)) e^{-V_N + o(\nu_N)}.$$

For clarity, before absorbing subpolynomial factors the profile term is

$$p^2 R_2(N, Q) \ll N^{-1/3 + \varepsilon} (\lambda^2 + \lambda 2^{E+2}).$$

If $I + \log^+ \lambda = O(V_N)$ and $E = O(V_N)$, multiplication by $e^I s(\lambda)$ leaves only an $N^{o(1)}$ factor. Start with a smaller arithmetic ε to obtain the final remainder $N^{-1/3 + \varepsilon}$ at any prescribed positive ε . This is the step that prevents an unrecorded exponential mark-cutoff loss.

For aggregation with signs retained, replace q_e by $q_{e,s} = q_e/2$ and $R_2(N, Q)$ by $R_2^{\text{val}}(N, Q+1)$ in (C.5). The local factor is at most four and $\sum_{e,s} \sqrt{q_{e,s}} = \sqrt{2}(1 + \sqrt{2})$, both fixed constants. The full-value profile at $Q+1$ vertices has the same envelope as the relative profile at Q rows. It therefore leaves the same polynomial remainder and the same one-factor budget. By contrast, the unweighted labelled argument sums signs and marks first and continues to use $R_2(N, L)$ at the base length.

C.4 Removing the mark cutoff

Assume $I + \log^+ \lambda \leq V_N - c\nu_N$, and choose $0 < c' < c'' < c$. Set

$$E_N = \left\lceil \frac{\log^+ \lambda + c'' \nu_N}{\log 2} \right\rceil.$$

Then $E_N = O(V_N) = o(\log N)$, and the target tail has mass $\mu_N = \lambda 2^{-E_N - 1} \leq e^{-c'' \nu_N}$. The actual tail event is exactly a hit at length $L + E_N + 1$. The scalar hard-conditioning inequality at that shifted length gives

$$\mathbb{P}(\exists x, e > E_N : K_{x,e} = 1 \mid C) \leq \mu_N + O\left(e^I \mu_N (e^{-V_N + o(\nu_N)} + N^{-1/3 + \varepsilon})\right).$$

It is $O(\mu_N)$ under the displayed budget. Couple each tail to zero, apply the finite theorem, and absorb $\log(1 + \log^+ \lambda) = o(\nu_N)$. This gives

$$d_{\text{TV}} \left(\mathcal{L}((X_e)_{e \geq 0} \mid C), \bigotimes_{e \geq 0} \text{Pois}(\lambda q_e) \right) \ll e^{-c' \nu_N} + N^{-1/3 + \varepsilon}.$$

For finite-mass sequences, tail summation is a measurable bijection, with inverse $X_e = T_e - T_{e+1}$. The entire threshold path has exactly the same total-variation distance.

C.5 Keeping the spatial labels

For the labelled comparison no factor (C.3) is inserted. The two simple identities

$$\sum_e p_e \leq p, \quad \sum_{e,f} \mathbb{P}(K_{x,e} K_{y,f} = 1) \leq \mathbb{P}(J_{x,L} J_{y,L} = 1)$$

then allow the separated-pair contribution to be folded at the *base* length. This is why the labelled finite inequality uses $R_2(N, L)$ even though its graph and bad supports use Q . At a

free cutoff $Y = e^w$ its leading errors after conditioning are

$$e^I \lambda e^{-D(H/w)+o(H/w)}, \quad e^I \lambda^2 e^{-w+o(H/w)}, \quad H = \log N.$$

For zero information cost the permitted $\log \lambda$ is below $\min\{D(H/w), w/2\}$. The first function decreases and the second increases, giving their crossing $w = \widehat{V}_N$. The sufficient budget $I + \log \lambda \leq \widehat{V}_N/2 - c\widehat{\nu}_N$ makes both errors tend to zero. It is a convenient sufficient condition; it is not asserted to be the exact optimized boundary for every positive information cost.

One can remove the marked tail by taking, for instance, $E = \lceil (I + \widehat{V}_N + \log(2 + \lambda)) / \log 2 \rceil$. On every nontrivial budget this is $O(\widehat{V}_N) = o(\log N)$. The unconditional first-moment bound at the shifted length makes the actual conditional tail $O(e^I \lambda 2^{-E})$, and the target tail is $O(\lambda 2^{-E})$; both are absorbed by the preceding ledger. If its right side is at least a constant, the distance bound follows instead from $d_{TV} \leq 1$, so no unjustified logarithmic-band choice is required outside the effective range.

C.6 Macroscopic transport and containment

Let $L_M = \lfloor \log_2 M \rfloor - a_M$, $a_M = o(\log M)$, and $\lambda = M 2^{-L_M}$. The base-contained interior starts are $2 \leq x \leq M - L_M + 1$; their number is $M - L_M$. First retain $\lceil M^{1/2} \rceil \leq x \leq M - L_M + 1$. The conditional ranks and graph construction are local and unchanged. The one-window defect masses, summed on dyadic slices, are $M^{1/2+o(1)}$; the edge estimate uses the upper height M ; and the separated-pair mass is bounded by the macroscopic profile (3.4). Thus the labelled finite comparison is

$$\ll \lambda(1 + \lambda)(e^{-V_M+o(\nu_M)} + M^{-1/3+\varepsilon}).$$

Restoring the deep starts costs the bound (7.7),

$$O\left(\lambda M^{-1/2} + \exp(-c_0 \log M / \log \log M)\right).$$

The target deletion costs only $O(\lambda M^{-1/2})$. Conditioning on a compatible event in the chosen prime field multiplies the averaged error by at most e^I .

Choose $E = \lceil (I + V_M + \log(2 + \lambda)) / \log 2 \rceil$ in the hard labelled regime. The exact mark-tail identity and the global scalar bound control the real tail. Besides the intensity-linear tail term, the scalar bound has the separate deep-start error just displayed; that error is already present in the full-field inequality. In particular it must not be silently dropped from a finite-prefix tail calculation. Both tails are absorbed by (7.14). The free-cutoff proof is identical with V_M replaced by the optimized cutoff in its two leading terms.

For aggregated counts apply Theorem C.1 on the retained macroscopic set, with $R_{2,\delta}(M, Q)$ in place of $R_2(N, Q)$. The same geometric square-root sum applies. Under $I + \log^+ \lambda \leq V_M - c\nu_M$, the deep-start correction times e^I is smaller than every fixed $e^{-c'\nu_M}$, and the aggregate mark cutoff above removes both tails. This proves the macroscopic one-factor staircase without an additional marked arithmetic input.

At exact centered level r the target mean is precisely $(M - L_M)2^{-b_M-r-1}$, not $M 2^{-b_M-r-1}$. All these fields retain the full rightward mark of a base-contained start. They are not the list of blocks censored at the right end of the finite sequence. The containment diagram in the article distinguishes the two objects.

C.7 Deterministic masks and diffuse limits

A deterministic restriction of a lattice field contracts total variation. At bounded intensity, if $N^{-1} \sum_{x \in A_N} \delta_{x/N} \Rightarrow \mu$, the restricted marked field therefore converges weakly to $\text{PPP}(\lambda \mu \otimes \nu_{\text{geo}})$,

along a subsequence $N2^{-L} \rightarrow \lambda$. The same statement holds for the macroscopic field, with scaling by M . It follows directly from the target Laplace functional and weak convergence of the deterministic empirical measures. There is no universal numerical rate for this last deterministic step without a quantitative assumption on the mask or the test function.

Nor is the diffuse limit a total-variation target. The actual field is supported on its lattice, while the diffuse target has positive probability $1 - e^{-\lambda\mu(\mathbb{R})}$ of a nonempty configuration outside that lattice. The approximation in total variation always takes place *before* passing from the grid to the continuum.

D Conditional laws and the Poisson target

The arithmetic theorems provide total-variation errors. This appendix records exactly when those errors are small enough to transfer a local probability, a conditional state, or a Gaussian limit. Properties of the independent Poisson target are not additional arithmetic results.

D.1 Scalar Gaussian and local estimates

Let $P_\lambda \sim \text{Pois}(\lambda)$ and let ϵ_N denote any proved conditional scalar total-variation bound in the article. If $\lambda \rightarrow \infty$,

$$\sup_t \left| \mathbb{P} \left(\frac{Z - \lambda}{\sqrt{\lambda}} \leq t \mid C \right) - \Phi(t) \right| \leq \epsilon_N + C\lambda^{-1/2}.$$

For example, write P_λ as a sum of $\lceil \lambda \rceil$ independent Poisson variables of mean at most one and use the classical Berry bound [5]. Their third absolute centered moments are $O(\lambda/\lceil \lambda \rceil)$, which gives the displayed scale. The normalization uses the deterministic comparison mean λ ; replacing it by an actual conditional mean requires a separate moment estimate and is not inferred from total variation alone.

For every integer $n \geq 1$ and every $\lambda > 0$, one has the exact form

$$\mathbb{P}(P_\lambda = n) = \frac{e^{-\delta_n}}{\sqrt{2\pi n}} \exp[-\lambda h(n/\lambda)], \quad 0 \leq \delta_n \leq \frac{1}{12n}. \quad (\text{D.1})$$

Here $h(u) = u \log u - u + 1$, and δ_n is independent of λ . To check the remainder, set

$$\delta_n = \log(n!) - (n + \tfrac{1}{2}) \log n + n - \tfrac{1}{2} \log(2\pi).$$

Stirling's limit gives $\delta_n \rightarrow 0$. With $t = (2n + 1)^{-1}$,

$$\begin{aligned} \delta_n - \delta_{n+1} &= (n + \tfrac{1}{2}) \log(1 + 1/n) - 1 = \sum_{j \geq 1} \frac{t^{2j}}{2j + 1}, \\ 0 \leq \delta_n - \delta_{n+1} &\leq \frac{t^2}{3(1 - t^2)} = \frac{1}{12n(n + 1)}. \end{aligned}$$

Telescoping to infinity proves the stated bounds. Substitution of the factorial identity into $e^{-\lambda} \lambda^n / n!$ proves (D.1); moreover $0 \leq 1 - e^{-\delta_n} \leq 1/(12n)$, uniformly in the intensity. For $n = 0$ the mass is exactly $e^{-\lambda}$ and the square-root formula is not used.

Thus relative transfer at n requires $\epsilon_N / \mathbb{P}(P_\lambda = n) \rightarrow 0$. For the hard scalar rate a sufficient condition is

$$I + \log^+ \lambda + \tfrac{1}{2} \log n + \lambda h(n/\lambda) \leq V_N - c\nu_N,$$

and for the soft rate a sufficient condition is

$$2I + \log^+ \lambda + \log n + 2\lambda h(n/\lambda) \leq \widehat{V}_N - c\widehat{\nu}_N.$$

In both cases the event must be measurable at the corresponding prime cutoff. The logarithms of the reciprocal target masses are then $O(V_N)$ or $O(\widehat{V}_N)$, so division by that mass does not destroy the polynomial remainder.

In the central range $n = \lambda + t\sqrt{\lambda} + O(1)$, $|t| = o(\lambda^{1/6})$, Taylor expansion gives $\lambda h(n/\lambda) = t^2/2 + o(1)$ and $\log n = \log \lambda + o(1)$. The respective central local budgets are therefore

$$I + \frac{3}{2} \log \lambda + \frac{1}{2} t^2 \leq V_N - c\nu_N, \quad 2I + 2 \log \lambda + t^2 \leq \widehat{V}_N - c\widehat{\nu}_N.$$

A global scalar approximation at its upper intensity endpoint does not by itself imply a relative local approximation there.

For completeness, the analogous relative upper-tail transfer has the additional denominator $\mathbb{P}(P_\lambda \geq \lceil \lambda + t\sqrt{\lambda} \rceil)$. Stirling's formula and summation of successive Poisson masses yield the usual Gaussian moderate-tail comparison for $0 \leq t = o(\lambda^{1/6})$. Equivalently, expanding the exponent uniformly on the contributing $O(\sqrt{\lambda}/(1+t))$ scale gives relative error $O((1+t^3)/\sqrt{\lambda})$; the exponentially smaller outer part of the sum is bounded by the ratio of successive masses. Since $1 - \Phi(t) \asymp e^{-t^2/2}/(1+t)$ for $t \geq 0$, a sufficient hard arithmetic condition is

$$I + \log^+ \lambda + \frac{1}{2} t^2 + \log(1+t) \leq V_N - c\nu_N.$$

This condition must be combined with the intrinsic Poisson restriction $t = o(\lambda^{1/6})$. It is not a claim about larger deviation scales.

D.2 Resolving an entire conditional path

Let ϵ^{path} be the aggregate staircase error based at the present threshold. The sharpened conditioning inequality (6.4) gives a full future-path error at most $\epsilon^{\text{path}}/p_\lambda(n)$, where $p_\lambda(n) = \mathbb{P}(P_\lambda = n) > \epsilon^{\text{path}}$. The same assertion holds jointly with any fixed available lower segment. Forward transitions are binomial thinning; backward transitions add independent Poisson increments.

Before division, the aggregate proof gives

$$\epsilon^{\text{path}} \ll e^I \lambda (1 + \log^+(2\lambda)) e^{-V_N + o(\nu_N)} + N^{-1/3+\varepsilon},$$

provided the information, log-intensity and mark cutoff are $O(V_N)$. Combining this with (D.1) proves the budget (6.7). Its hypotheses also validate that use of the common logarithmic band. Indeed, for $n \geq 2$, if $\lambda \geq 1$ the budget bounds $\log \lambda$ by $O(V_N)$. If $\lambda < 1$, use

$$\lambda h(n/\lambda) = n \log n - n + \lambda - n \log \lambda$$

to conclude $|\log \lambda| = O(V_N)$. Thus the base offset, information cost and chosen mark cutoff are all $O(V_N) = o(\log N)$. The atom's reciprocal is $\exp(O(V_N)) = N^{o(1)}$, which can be absorbed by first choosing a smaller arithmetic ε . For a fixed present state at a fixed positive limiting intensity the atom stays bounded away from zero, and no extra logarithmic cost is needed.

The condition $p_\lambda(n) > \epsilon^{\text{path}}$ only makes the conditional law well-defined and gives a bound. Convergence requires $\epsilon^{\text{path}}/p_\lambda(n) \rightarrow 0$. Nothing here supplies a single finite-intensity comparison for the entire reverse path: its target mass diverges toward thresholds $m \rightarrow -\infty$.

D.3 Quenched transfer along geometric scales

The proof of [Corollary 6.4](#) gives the single-length Markov–Borel–Cantelli argument. Its uniform extension uses the bound

$$e^{-(c-\beta)\nu_{N_k}+o(\nu_{N_k})} + e^{\beta\nu_{N_k}N_k^{-1/3+\varepsilon}}, \quad 0 < \beta < c,$$

for the exceptional probability on $N_k \asymp q^k$, $q > 1$. Since $\nu_{N_k} \asymp \sqrt{k/\log k}$, multiplication by $O(\log N_k) = O(k)$ preserves summability. A single union bound therefore covers every integer length in a fixed logarithmic band, provided the error margin is uniform. The resulting almost-sure statement concerns these conditional laws at individual scales; it requires no independence between scales and asserts no canonical coupling over all scales.

D.4 Exact target consequences

For a phase $\vartheta \in [0, 1]$, let $P_r \sim \text{Pois}(2^{\vartheta-r-1})$ independently for $r \in \mathbb{Z}$, and $S_m = \sum_{r \geq m} P_r$. Each tail sum is finite almost surely. Then

$$S_m \sim \text{Pois}(2^{\vartheta-m}), \quad \text{Cov}(S_m, S_n) = 2^{\vartheta-\max(m,n)}.$$

Splitting $S_m = P_m + S_{m+1}$ gives the exact forward thinning and backward immigration kernels in the article. For $s \geq 0$ the joint probability generating function is

$$\mathbb{E} \prod_{j=0}^s z_j^{S_{m+j}} = \exp \left[\sum_{r=m}^{m+s-1} 2^{\vartheta-r-1} \left(\prod_{j=0}^{r-m} z_j - 1 \right) + 2^{\vartheta-m-s} \left(\prod_{j=0}^s z_j - 1 \right) \right].$$

Conditionally on $S_m = n$, the n points have independent uniform positions and independent geometric overshoots $\mathbb{P}(E = e) = 2^{-e-1}$. For lattice targets the positions are uniform on the corresponding finite grid. Conditioning the critical prelimit field on a fixed count transfers this assertion by the sharp conditioning lemma; only the final grid-to-uniform passage is weak convergence.

The complete centered target is locally finite on $[1, 2] \times \mathbb{Z}$, although it has infinite total mass toward $r = -\infty$. For every nonnegative continuous compactly supported g put

$$\Psi_{\vartheta}(g) = \exp \left[- \int_1^2 \sum_{r \in \mathbb{Z}} 2^{\vartheta-r-1} (1 - e^{-g(t,r)}) dt \right].$$

Compact support meets finitely many levels. At those levels the lattice comparison and a Riemann sum give

$$\mathbb{E}(e^{-\langle g, \mathcal{E}_N \rangle} \mid C_N) = \Psi_{\vartheta_N}(g) + o_g(1)$$

whenever $I_N \leq V_N - c\nu_N$. A convergent phase subsequence is needed only to replace this moving deterministic target by a single limit law.

For a base intensity $\lambda_N \rightarrow \infty$, the exact maximum identity in the article, applied at $m_N = \lfloor \log_2 \lambda_N \rfloor + j$, gives, along $\{\log_2 \lambda_N\} \rightarrow \vartheta$,

$$\mathbb{P}(M_{N,L} - \lfloor \log_2 \lambda_N \rfloor \leq j) \longrightarrow e^{-2^{\vartheta-j-1}}.$$

This is a lattice extreme law. The identity uses the scalar comparison at the shifted critical length, not an unproved uniform law for every large marked field.

Finally, the Poisson–Gaussian bridge follows by splitting the independent levels at a fixed critical level r_0 . Below it, the normalized Poisson sums satisfy a multivariate central limit

theorem with covariance $2^{-\max(j,k)}$. The centered contribution of the levels at or above r_0 has variance $O(\Lambda_N^{-1})$ after normalization, whereas those levels retain their Poisson law and are independent of the lower part. This proves joint independence in the limit. Continuous-time embeddings and phase-removing auxiliary randomizations belong to the separate process article [11]; no such enlargement is used in the arithmetic proofs here.

E The microscopic boundary and relative factorization

The exact border cylinder and microscopic non-vacancy are different objects. The first has probability $2^{-\pi(L)}$ without an error term; the second also allows an interior start. This appendix proves that the entire interior first moment is exponentially smaller than the border probability. A union bound is then sufficient; no microscopic overlap expansion is needed.

E.1 Valuation matrices and the transition range

Put $B = L + 1$, $V_x = \{x - 1, \dots, x + L - 1\}$, and let M_x have rows $v(n) = (v_p(n) \bmod 2)_p$ for $n \in V_x$. The start matrix is $A_x = D_x M_x$, where D_x is the edge-difference matrix of a tree on V_x . Its kernel consists exactly of the constant vectors. Consequently

$$\text{rank } A_x \geq \text{rank } M_x - 1, \quad \mathbb{P}(J_{x,L} = 1) \leq 2^{-\text{rank } A_x}. \quad (\text{E.1})$$

For $2 \leq x \leq L - 1$ a start is impossible: both $2x$ and $2x - 2$ lie inside the putative constant run, and complete multiplicativity would force $f(2) = 1$ and $f(2) = -1$ simultaneously.

Lemma E.1 (Transition identity minor). *Uniformly for $x \in \{B - 1, B, B + 1\}$, $\text{rank } A_x \geq (2 - o(1))\pi(B)$.*

Proof. Write $a = x - 1 \in \{B - 2, B - 1, B\}$ and $Y_0 = 2\sqrt{B}$. For each prime $Y_0 < p \leq B$ choose $m_p = p[a/p] \in [a, a + B - 1]$. The cofactor satisfies $[a/p] \leq B/p + 1 < \sqrt{B}/2 + 1 < Y_0 < p$ for large B . Thus m_p has valuation one at p and no other prime factor above Y_0 . Each prime $B < q \leq a + B - 1$ occurs as its own vertex. On these rows and columns the valuation matrix has an identity minor of size $\pi(a + B - 1) - \pi(Y_0) = (2 - o(1))\pi(B)$. Apply (E.1). $\square$

E.2 The uniform prime-divisor input

The precise literature input is

$$\forall \varepsilon > 0 \exists B_0(\varepsilon) \forall B \geq B_0(\varepsilon) \forall n > B : \quad \omega \left(\prod_{j=0}^{B-1} (n + j) \right) \geq (2 - \varepsilon)\pi(B). \quad (\text{E.2})$$

It is equation (14) of Laishram–Shorey [9]. Its uniformity is important. For $n > 17B/12$, their Theorem 2 gives at least $\pi(2B)$ prime divisors, apart from a fixed exceptional pair. For $B < n \leq 17B/12$, every prime at most B divides the block product, and each prime in $[n, n + B - 1]$ is an additional divisor. Hence the number is at least

$$\pi(B) + \pi(n + B - 1) - \pi(n - 1) = (2 - o(1))\pi(B),$$

uniformly in that bounded ratio, by the prime number theorem. This verifies the quantifier order in (E.2).

Suppose now that $B + 2 \leq x \leq B^2 + 2$. Every vertex is at most $B^2 + B$. Equation (E.2) supplies $t_x \geq (1 - o(1))\pi(B)$ primes $q > B$ in the block. Each has valuation one, since

$q^2 \geq (B+1)^2 > B^2 + B$; it occurs at one vertex only, since $q > B-1$; and two such primes cannot occupy the same vertex, since their product exceeds $B^2 + B$. Their columns span a coordinate subspace U_x of dimension t_x . Call its rows the private rows.

E.3 A global count of even valuations

Fix an integer $K \geq 2$ and use the medium primes $B/K < p \leq B$. Let E_x be their total odd-valuation incidence mass in V_x . We claim

$$E_x \geq (H_K - 1 - o(1))\pi(B), \quad H_K = \sum_{j=1}^K \frac{1}{j}. \quad (\text{E.3})$$

The block contains at least $\lfloor B/p \rfloor$ multiples of p . An incidence of positive even valuation is among those with $n = ap^2 \leq B^2 + B$. Thus $a \leq K^2 + 1$ for sufficiently large B . For fixed a , successive distinct primes $p, q > B/K$ give values ap^2, aq^2 separated by more than $2aB/K$; only $1 + K/(2a)$ such values can lie in an interval of length B . Summing over the fixed number of possible a shows that the total number of square divisibilities is $O_K(1)$, uniformly in x . Subtracting all of them is a permissible overcount of even valuations.

The main incidence count can be evaluated without rounding every summand:

$$\begin{aligned} \sum_{B/K < p \leq B} \left\lfloor \frac{B}{p} \right\rfloor &= \sum_{j=1}^{K-1} (\pi(B/j) - \pi(B/K)) \\ &= (H_{K-1} - (K-1)/K + o(1))\pi(B) \\ &= (H_K - 1 + o(1))\pi(B). \end{aligned}$$

This proves (E.3). The improvement over a prime-by-prime loss is essential: subtracting one for each prime would lose a positive proportion of $\pi(B)$.

E.4 Rank of a simple incidence matrix

Lemma E.2 (Simple incidence rank). *Let H be a finite binary matrix. Suppose every row has weight at most two, every column has weight at most K , and no pair of columns occurs as the support of two weight-two rows. If E is the number of entries equal to one, then*

$$\text{rank } H \geq \frac{E}{K+1}.$$

Proof. Remove zero columns and view a weight-two row as an edge between its columns, and a weight-one row as a pin. In a connected component with r active columns, an unpinned component has rank $r-1$ and a pinned component has rank r . The first assertion follows because the edge vectors of a connected graph span the even-parity subspace; one pin then spans its missing direction.

A pinned component has incidence mass at most Kr , which is at most $(K+1)$ times its rank. An unpinned component has $r \geq 2$; simplicity and the degree bound give incidence mass at most $\min\{r(r-1), Kr\}$. If $r \leq K+1$ use the first bound, and if $r \geq K+1$ use the second. In either case the mass is at most $(K+1)(r-1)$. Sum over the components. $\square$

Delete the private rows from the medium-prime columns of M_x , and call the resulting matrix H_x . For fixed K and sufficiently large B , a private row carries at most one medium prime: otherwise a vertex would exceed $B(B/K)^2 > B^2 + B$. Thus deletion removes at most t_x incidences. A remaining row carries at most two medium primes, since $(B/K)^3 > B^2 + B$. A

column has weight at most K . Finally two weight-two rows with the same support p, q would be two multiples of $pq > (B/K)^2 > B$ inside one block, which is impossible. All hypotheses of Lemma E.2 are therefore met. Modulo U_x , a medium column is represented by its restriction to the remaining rows, so

$$\begin{aligned} \text{rank } A_x &\geq t_x + \frac{(E_x - t_x)_+}{K+1} - 1 \\ &\geq \left(1 + \frac{H_K - 2}{K+1} - o(1)\right) \pi(B). \end{aligned} \quad (\text{E.4})$$

For the last step, if $E_x \leq t_x$ use $t_x \geq E_x$; otherwise use $t_x \geq (1 - o(1))\pi(B)$ and (E.3). The parameter can be chosen exactly within this family. Set $\delta_K = (H_K - 2)/(K + 1)$. Then

$$\delta_{K+1} - \delta_K = \frac{3 - H_K}{(K+1)(K+2)}.$$

Since $H_{10} < 3 < H_{11}$ and H_K is strictly increasing, the unique maximizer over integers $K \geq 2$ is $K = 11$. Consequently we use

$$H_{11} = \frac{83711}{27720}, \quad \delta_* := \frac{H_{11} - 2}{12} = \frac{28271}{332640} > \frac{1}{12}.$$

This optimizes only the displayed fixed-cutoff family of bounds, not the rank surplus among all possible methods.

The local product comparisons hold for $B \geq 11^3 + 1 = 1332$: $(B/11)^3 > B^2 + B$, $B^3/121 > B^2 + B$ and $(B/11)^2 > B$. In the square-divisibility count one may then use $a \leq 122$ and at most seven values per a , giving the safe bound 854. These elementary checks do *not* make 1332 a threshold for the complete asymptotic theorem: the prime-divisor and prime number theorem thresholds still depend on the desired error margin.

E.5 Post-quadratic starts and summation

Lemma E.3 (Post-quadratic defect maximum). *There is an absolute constant θ_0 such that, for all sufficiently large B and every $x > B^2 + 2$, the number m_x of B -defective vertices in V_x is at most $B - g_L$, where*

$$g_L = \frac{B}{\log B} (\log \log B - \log \log \log B - \theta_0), \quad g_L / \pi(B) \longrightarrow \infty.$$

In particular $\mathbb{P}(J_{x,L} = 1) \leq 2^{-g_L}$.

Proof. We use the Balasubramanian–Shorey exclusion [3] in the form recorded by Shorey [13, pp. 55–56, equation (15) with exponent 2 and the subsequent square-case statement]. It has an absolute constant θ_0 and, for every fixed $\varepsilon > 0$, a threshold $k_0(\varepsilon)$ such that, for $k \geq k_0(\varepsilon)$,

$$\prod_{i=1}^t (z + d'_i) = by^2, \quad 0 \leq d'_1 < \cdots < d'_t < k, \quad b > 0, \quad P^+(b) \leq k, \quad z > e^{1-\theta_0+\varepsilon} \frac{k \log k}{\log \log k}$$

forces $t < \mu_k(\theta_0)$, where

$$\mu_k(\theta) = k \left(1 - \frac{\log \log k}{\log k} + \frac{\log \log \log k}{\log k} + \frac{\theta}{\log k} \right).$$

The smoothness condition on b is part of (15); it is not a size bound on b . All thresholds are uniform in z, b, y and the shifts. To obtain the form used here, first fix $\varepsilon = 1$. Increase the absolute threshold until

$$k^2 > e^{2-\theta_0} \frac{k \log k}{\log \log k}.$$

For $m > k^2$ and $1 \leq d_1 < \dots < d_t \leq k$, set $z = m + 1$ and $d'_i = d_i - 1$. These substitutions preserve every factor $m + d_i = z + d'_i$, put all shifts in $[0, k]$, and satisfy the required height bound. Hence the product equation forces $t < \mu_k(\theta_0)$. Write $V_x = \{m + 1, \dots, m + B\}$ with $m = x - 2 > B^2$. The product of its B -defective vertices is a square times a B -smooth integer. There is no extra upper bound on that smooth coefficient in the imported statement. Its contrapositive gives $m_x < \mu_B(\theta_0) = B - g_L$ beyond the fixed absolute threshold. The affine defect bound is $\mathbb{P}(J_{x,L} = 1) \leq 2^{-L} 2^{\max(m_x-1, 0)}$, which is at most 2^{-g_L} for large B . $\square$

Partition the starts $2 \leq x \leq 2L^2$ into the forbidden early range, the three transition starts, $B + 2 \leq x \leq B^2 + 2$, and the remaining post-quadratic starts. Their total probabilities are respectively

$$0, \quad 3 \cdot 2^{-(2-o(1))\pi(B)}, \quad O(B^2) 2^{-(1+\delta_*-o(1))\pi(B)}, \quad O(L^2) 2^{-g_L}.$$

Since $\log B = o(\pi(B))$ and $\pi(B) = \pi(L) + O(1)$, their sum is at most $2^{-(1+\delta_*-o(1))\pi(L)}$. Adding the exact border event proves

$$q_L = 2^{-\pi(L)} (1 + O_{\delta_0}(2^{-\delta_0 \pi(L)})) \quad (0 < \delta_0 < 28271/332640),$$

including the displayed choice $\delta_0 = 1/12$. The same interior sum divided by q_L bounds the conditional probability that the exact border is not the unique microscopic start.

The cutoff-invariance and mesoscopic estimates use one further observation. At heights above $2L^2$, two B -defective vertices at distance at most L have distinct squarefree B -smooth parts; equal parts would place consecutive squares too far apart. The number of choices of these parts is $\exp(O(L/\log L))$, and the uniform Pell bound gives the same type of bound on the number of corresponding windows in the entire admissible population up to twice the ambient height. Multiplication by 2^{-g_L} remains $o(2^{-\pi(L)})$, without an exterior height-slice factor. Starts with at most one defect cost only their number times 2^{-L} . This proves the bounds in the article with cutoff $K \leq 2^L$, using 2^L as an artificial upper ambient scale; it does not require a dyadic lower endpoint comparable to that artificial scale.

E.6 The prime clock

Section 7.5 of the article proves the exact prime-clock law and explains why integer overshoots need not be tight. We record here the conventions needed for the conditional comparisons. Let $J_\partial = \min\{j : f(p_j) = -1\}$. On $C_L = 1$ and the full-measure event $J_\partial < \infty$, put $G_L = J_\partial - \pi(L) - 1$; assign $G_L = 0$ elsewhere. Then

$$\mathbb{P}(G_L = k \mid C_L = 1) = 2^{-k-1}, \quad \mathbb{P}(\ell_\partial - L \geq h \mid C_L = 1) = 2^{-[\pi(L+h)-\pi(L)]}.$$

Since $\{C_L = 1\} \subseteq \{B_L^{\text{mic}} > 0\}$, the conditional law of G_L given microscopic non-vacancy is a mixture whose $C_L = 1$ component is exactly geometric. Its distance from that law is at most $\mathbb{P}(C_L = 0 \mid B_L^{\text{mic}} > 0) = O(2^{-\delta_0 \pi(L)})$. The next subsection constructs the finite-cylinder representative used to carry this mark through the stable product comparison.

E.7 Relative bulk estimates and affine conditioning

Let $\lambda = M2^{-L} \rightarrow 0$, $L = O(\log M)$ and $U = [\lceil M^\delta \rceil, M - L + 1] \cap \mathbb{Z}$. Fix δ and an upper bound for $L/\log M$. With $Y = Y_M^\sharp$, truncate the signed exact marks at E and put $Q = L + E + 1$. The averaged conditional process error is at most

$$C \left[p\mathcal{M}_B + p|D_Y(Q)| + p^2(MQ + E_Y(Q) + R_{2,\delta}(M, L)) \right].$$

The local ranks are those in Appendix C. Summing signs and exact marks before the separated-pair estimate leaves the relation sum at the base length L ; only the bad supports and shared-prime graph use Q . After division by $Mp = \lambda$, the contributions are

Source	Normalized contribution
One-window defects	$M^{-1/2+o(1)}$
Bad maximal supports	$e^{-V_M+o(\nu_M)}$
Diagonal and local pairs	pQ
Shared-prime edges	$\lambda(e^{-V_M+o(\nu_M)} + M^{-1+o(1)})$
Base-length relations	$M^{-1/3+\varepsilon}(1 + \lambda)$
Actual and target mark tails	$O(2^{-E-1})(1 + M^{-1/2+o(1)})$

For the last row, the exact tail event is a start at length $L + E + 1$; the masked first-moment estimate at that length gives the displayed bound after averaging the conditional tail probability. Choose $E = \lceil V_M/\log 2 \rceil$. Then $Q = O(\log M)$, $pQ = O(\log M/M)$ for large M , and every row tends to zero. This proves the signed, marked relative error in (7.15), not merely its unmarked projection.

The microscopic vector is measurable at the same prime cutoff because $2L^2 + L < Y$ for large M . For a fixed K , define $G_{L,K}$ from the first K prime signs after L : count consecutive positive signs, capped at K , and put the value zero off $C_L = 1$. This finite-cylinder variable agrees almost surely with $G_L \wedge K$; this makes measurability on the finite prime sigma-field explicit, independently of the convention on the null nonterminating event. Bertrand's postulate bounds the K th prime after L by $2^K L < Y$ eventually, for fixed K . The stable lift retains $(\mathbf{B}_L, G_{L,K})$ jointly with the full signed marked bulk field, at cost $\Delta_M = o(\lambda)$.

Write $\alpha = 2^{-\pi(L)}$, $b = |U|p$, and let u_M and v_M denote the interior microscopic first moment and the mesoscopic hit probability. The source bounds give $u_M + v_M = o(\alpha + b)$. The one-hit calculation in the proof of Theorem 7.9 gives sparse-target mass $e^{-b}[\alpha + (1 - \alpha)b]$ and a change of mixture weights at most b relative to (7.22). The conditional bulk mark, sign and position retain their full laws; only the boundary clock is truncated. Simultaneous sources and multiple bulk points have probabilities at most αb and $b^2/2$. Since the sparse-event mass is asymptotic to $\alpha + b$, normalizing the source subprobability measures gives

$$d_{\text{TV}}(\mathcal{L}(\Gamma_{M,L}^{[K]} \mid R_M \geq L), Q_{M,L}^{[K]}) \ll \frac{\Delta_M + u_M + v_M + \alpha b + b^2}{\alpha + b} + b.$$

The additional b is already absorbed by the last two numerator terms. For the actual first-hit law the discarded boundary tail is at most 2^{-K-1} : its unnormalized mass is at most $\alpha 2^{-K-1}$ and the hit probability is at least α . The target tail is also at most 2^{-K-1} . First let $M \rightarrow \infty$, then let $K \rightarrow \infty$. No lower bound on either mixture weight, convergence of their ratio, or exact finite-size independence is required. The grid-to-Lebesgue passage is a separate weak limit.

Now let $A = \{GX = b\}$ be a compatible affine cylinder on the exposed prime signs. Write $r = \text{rank } G$, $I = r \log 2$, and $U_L = \text{span}\{e_p : p \leq L\}$. If the stack with C_L is compatible, its

rank is $r + \pi(L) - \kappa$, where $\kappa = \dim(\text{row}(G) \cap U_L)$. Consequently

$$\mathbb{P}(C_L = 1 \mid A) = 2^{-(\pi(L) - \kappa)}$$

exactly, whenever the exposed cutoff contains the border coordinates. This finite identity needs no rare-intensity regime. If the stack is incompatible, that probability is zero. The parameter κ counts consequences entirely supported on the border coordinates, not input equations that merely mention them.

For the asymptotic conclusions assume $M, L \rightarrow \infty$, $L \leq A_0 \log M$, $M2^{-L} \rightarrow 0$, and eventually $I \leq V_M - c\nu_M$ for fixed $c > 0$. The matrices, their dimensions and right-hand sides may vary with M . Multiply the quantitative normalized bulk ledger by e^I before taking limits. The cutoff term is at most $e^{-c\nu_M + o(\nu_M)}$; all polynomial terms still vanish because $I = O(V_M) = o(\log M)$. Thus the amplified bulk error is $o(\lambda)$; amplification of an unspecified little-oh alone would not justify this conclusion. Also $I = O(V_M) = o(\pi(L))$ and $\kappa \leq r$. The conditioned microscopic interior mass, relative to $2^{-\pi(L) + \kappa}$, is at most

$$\exp(I)2^{-\delta_0\pi(L) - \kappa} = o(1).$$

The mesoscopic first term is $\lambda M^{\delta-1}$ and remains $o(\lambda)$ after multiplication by e^I ; its other term is dominated by 2^{-g_L} and remains negligible relative to the border mass. The same one-hit calculation, normalized by $2^{-\pi(L) + \kappa} + b$, therefore proves the affine-conditioned crossover with $\pi(L)$ replaced by $\pi(L) - \kappa$. No assumption on future prime coordinates is used for the rare tail, border concentration or location laws.

Finally let $q_1 < \dots < q_K$ be the first primes after L , and put $Q_{L,K} = \text{span}\{e_{q_1}, \dots, e_{q_K}\}$ and $H_L = \text{row}(G) + U_L$. If the extended stack is compatible,

$$\mathbb{P}(G_L \geq K \mid A, C_L = 1) = 2^{-[K - \dim(H_L \cap Q_{L,K})]};$$

otherwise this tail probability is zero. Thus the ordinary geometric mark is preserved when, for every fixed K , the condition $H_L \cap Q_{L,K} = \{0\}$ holds beyond a threshold that may depend on K . This is separate from the parameter κ controlling the border mass. Fix K , use the finite-cylinder representative and joint stable lift, and take the size limit before $K \rightarrow \infty$. No common threshold in K and no neutrality for a growing truncation $K(M)$ is assumed. The exact rank formula remains the alternative when neutrality fails; no relative estimate for a disappearing mixture component is inferred.

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